SANTOSH PANT

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EDUCATION

Knox College **September 2023 - June 2027**
Bachelor of Science in Computer Science and Data Science (GPA: 3.82) *Galesburg, Illinois*

- **Hackathon Wins:** HackIllinois 2026 (UIUC) - Best Use of Actian VectorAI DB (1st, sponsor track); UMich AI for Heat Resilience - 2nd Place ($2K); ASA DataFest 2026 - Best Insights Track; HackAugie 2026 - Best Gaming Hack
- **Honors:** Break Through Tech AI Fellow (Cornell Tech, 2025-26); Richter Research Scholar; Dean's List (6/6 terms); ICPC North Central Participant (UIUC site, 2025); Teaching Assistant - Data Structures & Algorithms; CS Club Lead

EXPERIENCE

Machine Learning Engineer - Research Collaborator **June 2026 - Present**
University of Michigan - Center for Global Health Equity (Mentor: Prof. Geoffrey Siwo) *Remote*

- Predicting street-level urban heat maps from RGB drone imagery (no thermal camera), toward cheap, scalable urban heat-risk mapping.
- Diagnosed two silent data faults the hackathon pipeline missed (colormapped-JPEG values used as targets instead of radiometric temperature; RGB/thermal pairs unaligned from mismatched camera fields of view) and rebuilt the pipeline (ConvNeXt U-Net, pix2pix cGAN); the 2nd-place hackathon model scored 19.3 dB PSNR / 0.66 SSIM on a 202-image held-out set.

Software Engineering Intern - The MACRO Consortium **March 2025 - June 2025**

- Designed and shipped a priority-based scheduling algorithm into the open-source pyscope package; resolved resource conflicts and improved allocation fairness for faculty observation requests in high-demand slots.
- Investigated and mitigated a data-loss bug in the scheduler's request-handling logic, materially improving reliability for nightly observation batches used at Knox College and partner institutions; wrote a pytest suite and refactored legacy parsing code to prevent regressions.

Software Developer Intern - Synapse Technologies **June 2024 - August 2024**

- Built an internal full-stack stock market simulation platform (React, Express.js, PostgreSQL) for backtesting and evaluating quantitative trading strategies.
- Designed schemas and APIs for real-time data ingestion and P&L visualization across 50+ simulated strategy instances; improved backend performance with Redis caching and validated load characteristics with k6.

PROJECTS

OMS - Crypto Order-Matching Engine | Go, PostgreSQL, React, Docker, Prometheus theoms.vercel.app

- Built a from-scratch in-memory limit-order-book matching engine in Go (price-time priority, partial fills, IOC/FOK/post-only, self-trade prevention, exact integer-money math); benchmarked at ~4.2M orders/sec.
- Wrapped it in a full trading platform - pre-trade risk, average-cost P&L, event sourcing with deterministic replay, TWAP/VWAP/Iceberg execution algos, a FIX 4.4 gateway, live WebSocket market data, and Prometheus metrics - plus a React trading-terminal UI, one-command Docker stack, CI, and 60+ tests.

Vigilant AI | FastAPI, Modal A100, YOLO11, CLIP ViT-L/14, Qwen2.5-VL, React **HackIllinois 2026 - 1st (sponsor track)**

- Built an AI system in 48 hours (two-person team) that turns raw CCTV footage into structured, searchable intelligence with no manual review; won its sponsor track at HackIllinois 2026 (github.com/thesantoshpant/vigilant-ai).
- Ran object detection (YOLO11), CLIP embedding search, and a vision-language model (Qwen2.5-VL) in parallel on a serverless A100; delivered hybrid CLIP + SQLite semantic search, plain-English chat over footage with SSE streaming, and one-click PDF incident export.

Yaar | React, TypeScript, Node.js, Express, MongoDB, Gemini (Vertex AI) okyaar.vercel.app

- Built a full-stack AI study-abroad counselor: a multi-agent pipeline behind a human-approval gateway, persistent per-student memory, Gemini-graded mock IELTS/TOEFL, Google OAuth + JWT, rate limiting, and dollar-cap / kill-switch spend controls.

bhasha-js | TypeScript, React, npm npmjs.com/package/bhasha-js

- Published a drop-in internationalization (i18n) SDK to npm that intercepts website text at the component level and serves AI-generated translations, requiring zero per-language developer work.
- Designed a lazy-translation model: untranslated strings are translated on first request, cached for subsequent visitors, and auto-refresh when the source text changes.

SKILLS

- **Languages:** Python, Go, TypeScript, JavaScript, Java, SQL
- **Backend & Cloud:** Node.js, Express.js, Flask, FastAPI, PostgreSQL, MongoDB, Redis, AWS, Docker, CI/CD
- **Frontend:** React, Next.js, Tailwind CSS
- **ML/AI & Testing:** PyTorch, CLIP/YOLO, LLM APIs, Git, Linux, pytest, Playwright, Jest